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Bihar Engineering University, Patna

B.Tech 2nd Semester Examination - 2026

Time: 03 Hours

Code: 100211 ***Subject: Workshop*** ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct option / answer the following (Any seven question only): **[2×7=14]**

(a) Which tool is used for marking lines on sheet metal?
(i) Scriber (iii) Snips
(ii) Punch (iv) Pliers

(b) Which of the following is a product developed in sheet metal practice?
(i) Spur Gear (iii) Pulley
(ii) Hexagonal Box with Cap (iv) Shaft

(c) Gas welding commonly uses
(i) Oxygen and Acetylene (iii) LPG only
(ii) Oxygen and Nitrogen (iv) Air only

(d) Which joining process melts the base metal?
(i) Brazing (iii) Welding
(ii) Soldering (iv) Riveting

(e) Which joining method is permanent?
(i) Nut and Bolt (iii) Screw Joint
(ii) Welding (iv) Cotter Joint

(f) A split pattern consists of
(i) One Piece (iii) Three Pieces
(ii) Two Pieces (iv) Four Pieces

(g) Which tool is used to hold the work piece during fitting?
(i) Bench Vice (iii) Pliers
(ii) Hammer (iv) Punch

(h) Which of the following is a hot working process?
(i) Rolling (iii) Filing
(ii) Grinding (iv) Turning

(i) Seasoning of wood is done to
(i) Increase moisture (iii) Increase weight
(ii) Remove moisture (iv) Change colour

(j) Which measuring tool gives the highest accuracy?
(i) Steel Rule (iii) Micrometer
(ii) Vernier Caliper (iv) Divider

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Q.2

(a) Describe the construction, working principle, and applications of the following sheet metal tools: (i) Steel Rule (ii) Vernier Caliper (iii) Micrometer (iv) Sheet Metal Gauge (v) Scriber and Divider [7]

(b) Explain the various operations performed in sheet metal working such as scribing, bending, shearing, punching, etc. Describe the procedure, tools used, and precautions for each operation with the help of sketches. [7]

Q.3

(a) Explain the Arc Welding process with a neat sketch. Discuss its principle, equipment, types of electrodes, advantages, and disadvantages. [7]

(b) Describe the soldering process with a neat sketch. Compare soft soldering and hard soldering, and mention their applications. [7]

Q.4

(a) List and explain various pattern materials used in foundry practice. Discuss their properties, advantages, limitations, and suitable applications. [7]

(b) What are the different types of moulding sands used in foundry? Explain its properties, composition, and applications. Why are additives used in moulding sand? [7]

Q.5

(a) Discuss the various safety measures and precautions to be observed in the Fitting Shop. Explain common hazards, personal protective equipment and good housekeeping practices. Why is safety important in workshop practice? [7]

(b) Classify files used in fitting shop according to their cut, shape, and grade. Explain the construction, uses and specification of different types of files with neat sketches. [7]

Q.6

(a) Define carpentry and explain its importance in engineering workshop practice. Classify the types of wood and discuss their properties, advantages, and disadvantages. [7]

(b) What is seasoning of wood? Explain the necessity of seasoning. Describe the different seasoning methods (natural and artificial) and discuss the common defects in wood with their causes and effects. [7]

Q.7

(a) Give a brief introduction to the Smithy Shop. Explain the differences between hot working and cold working processes. Discuss the advantages and limitations of each. [7]

(b) What is forging? Explain the classification of forging processes. Discuss the advantages of forging over casting and machining. [7]

Q.8

(a) With neat sketches, explain the construction and use of the following forging tools: (i) Anvil and Swage block (ii) Different types of Tongs (iii) Set Hammer and Flatter. [7]

(b) Describe the construction and working of a Wood Working Lathe with a neat sketch. Explain the common operations performed on it. [7]

Q.9 Write short note on any two of the following: **[7×2=14]**

(a) Working of Oxy-Acetylene Welding process with a neat sketch

(b) Carpentry joints with neat sketches and applications

(c) Match Plate patterns and Gated patterns with neat sketches and applications

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(d) Functions of the following foundry tools with neat sketches: Rammer, Trowel, Slick, Lifter, Sprue pin, Bellow, Mallet and Vent rod

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